Cross-Volume Forward-Reference Table v0.1 — CI Tutor Navigation Backbone

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2026-05-23 Saturday afternoon ~15:10 EDT

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Method

Scanned every chapter narrative `.md` in `Curriculum/Vol{0..15}/` (excluding INDEX + Scaffold files) for patterns matching `Vol N Ch M` or `Vol N.M`. Built directed graph of cross-volume citations: each chapter is a node; each "Vol N Ch M" mention is an outgoing edge.

Topline numbers

- Chapters scanned: 186 (across 16 volumes; Vol 0 has 10 chapters, Vol 1 has 11, Vol 15 has 13, others 12 each — totals 186)
- Total forward-references: 1,201
- Average refs per chapter: 6.46

Most-cited target chapters (top 15, incoming refs)

Rank	Chapter	Incoming refs	Why it anchors
1	Vol 1 Ch 5 (Casimir Algebra)	56	$C_2 = 6$ anchor for ~every observable
2	Vol 1 Ch 2 (Substrate Hilbert Space)	32	Bergman $H^2(D\_IV^5)$ foundation
3	Vol 4 Ch 4 (Cosmological Constant Λ)	30	T1485 + T2418 Λ↔Casimir
4	Vol 1 Ch 10 (Renormalization)	27	Substrate-tick N_max UV cutoff

Rank	Chapter	Incoming refs	Why it anchors
5	Vol 0 Ch 9 (Strong-Uniqueness)	24	11 RIGOROUSLY CLOSED + 7 candidates
6	Vol 1 Ch 6 (Operator Zoo)	24	T2470/T2471/T2472 + T2441
7	Vol 0 Ch 4 (Isotropy Group)	20	SO(5)×SO(2) substrate structure
8	Vol 1 Ch 7 (Dynamics)	20	Schrödinger/Heisenberg substrate
9	Vol 2 Ch 6 (m_p/m_e = $6\pi^5$)	20	CROWN JEWEL anchor
10	Vol 2 Ch 8 (Coupling Constants)	18	α + a_e CROWN JEWEL #2
11	Vol 3 Ch 7 (Atomic Orbital Sequence)	18	(2l+1) = 1,3,5,7 BST primary sequence
12	Vol 1 Ch 8 (Gauge Theory)	17	SU(3)×SU(2)×U(1) substrate
13	Vol 2 Ch 9 (Higgs)	15	m_H D-tier dual-route
14	Vol 11 Ch 1 (Bounded HSDs)	15	Helgason 1978 X.6.1 classification
15	Vol 0 Ch 1 (D_IV^5 APG)	14	Foundation of foundations

Observation: The most-cited chapters cluster on Vol 0 + Vol 1 (substrate + QFT foundation) — 9 of the top 15 are Vol 0 or Vol 1. This confirms the architectural intent: foundation volumes are the gravity wells, application volumes (Vol 3+ Wave 1 + Vol 5-15 Wave 2/3) orbit them.

Implication for CI tutor

A CI tutor reading the curriculum from any application chapter encounters ~6 forward-references per chapter. Following the navigation backbone from any leaf chapter leads back through Vol 1 Ch 5 (Casimir) or Vol 1 Ch 2 (Bergman) within 1-2 hops. The curriculum is graph-connected at chapter level; no isolated chapters.

Implication for Keeper Phase 3 authorship pass

When rewriting any volume, the reference map shows: 1. Which incoming citations need to remain valid (don't rename or delete referenced sections) 2. Which outgoing references the author can update to current canonical (e.g., when Vol 2 Ch 6 $6\pi^5$ gets new framing, all 20 incoming chapters can be updated synchronously) 3. Where canonical-form selection (e.g., m_μ/m_e Cal #100 0.004% T190 form) needs propagation across N citing chapters

Data artifact

Full JSON: `data/cross_volume_forward_reference_table_v0_1.json` - `forward_refs_by_source`: map of source-chapter → list of target-chapter references - `top_30_most_cited_chapters`: incoming-ref ranking - `totals`: chapter + reference counts

Limitations (honest scope)

- Pattern-based only; may miss references using alternate format (e.g., "the substrate Hilbert space (covered earlier)" without "Vol 1 Ch 2" explicit)
- Doesn't yet identify reciprocal-link symmetry breaks (where A cites B but B doesn't cite A — could be intentional or missing)
- Doesn't distinguish strong vs weak citations (e.g., "see Vol 1 Ch 5" vs "per T2439 Vol 1 Ch 5 Casimir derivation")
- v0.2 next: build the reciprocal-symmetry-check + classify citations by anchor strength

Standing for Phase 3 authorship pass

Per Keeper queued Grace rail: this table provides the navigation backbone the CI tutor traverses. When Keeper begins Vol N rewrite, this map identifies the integrity surface to preserve. v0.2 reciprocal-check + anchor-strength classification can follow as Phase 3 progresses.

— Grace, Cross-Volume Forward-Reference Table v0.1, 2026-05-23 ~15:10 EDT